IT5437 Computer Vision - Assignment 02: Fitting and Alignment

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Q1 - Line Fitting

(a) Total Least Squares - Line 1

TLS minimises perpendicular (orthogonal) distances from points to the line. The x1, y1 data is centred, then SVD applied to the 2-column data matrix. The right singular vector of the smallest singular value gives unit normal [a, b].

```
_, _, Vt = np.linalg.svd(np.column_stack([x1-mx, y1-my]))
a, b = Vt[-1];    c = -(a*mx + b*my)
```

https://github.com/PathmikaW/cv-assignment02/blob/main/src/q1_line_fitting.py

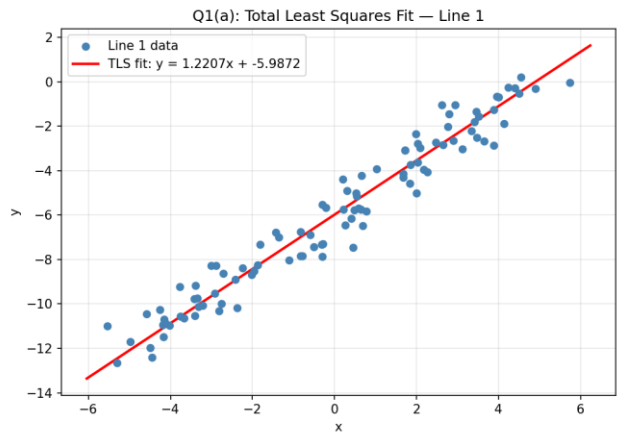
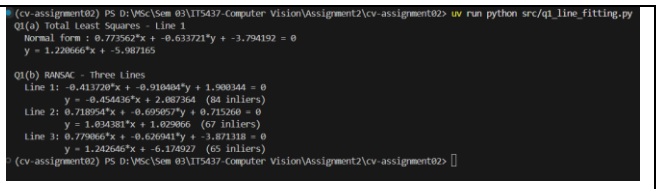


Fig 1: TLS fit - Line 1



```
(cv-assignment02) PS D:\MSc\Sem 03\IT5437-Computer Vision\Assignment2\cv-assignment02> uv run python src/q1_line_fitting.py
Q1(a) Total Least Squares - Line 1
  Normal form : 0.773562*x + -0.633721*y + -3.794192 = 0
  y = 1.220666*x + -5.987165

Q1(b) RANSAC - Three Lines
  Line 1: -0.413720*x + -0.910404*y + 1.900344 = 0
          y = -0.454436*x + 2.087364  (84 inliers)
  Line 2: 0.718954*x + -0.695057*y + 0.715260 = 0
          y = 1.034381*x + 1.029066  (67 inliers)
  Line 3: 0.779066*x + -0.626941*y + -3.871318 = 0
          y = 1.242646*x + -6.174927  (65 inliers)
(cv-assignment02) PS D:\MSc\Sem 03\IT5437-Computer Vision\Assignment2\cv-assignment02>
```

Terminal output - Q1

Singular values: [45.538, 5.014] - large ratio confirms strong linear structure.

Normal form: 0.7736x - 0.6337y - 3.7942 = 0

Slope-intercept: y = 1.2207x - 5.9872 (mean perpendicular distance = 0.406 units)

(b) RANSAC - Three Lines

All 300 points (X_all, Y_all flattened from all 6 columns) processed with sequential RANSAC (2000 iterations, perpendicular threshold = 0.5). Each round: sample 2 points, fit TLS line, count inliers. Best line kept, its inliers removed, repeat for remaining points.

```
X_all = D[:, :3].flatten();    Y_all = D[:, 3:].flatten()
```

https://github.com/PathmikaW/cv-assignment02/blob/main/src/q1_line_fitting.py

Line 1: -0.4137x - 0.9104y + 1.9003 = 0 => y = -0.4544x + 2.0874 (84 inliers)

Line 2: 0.7190x - 0.6951y + 0.7153 = 0 => y = 1.0344x + 1.0291 (67 inliers)

Line 3: 0.7791x - 0.6269y - 3.8713 = 0 => y = 1.2426x - 6.1749 (65 inliers)

Line 3 closely matches Q1(a) TLS (y = 1.2207x - 5.9872, slope diff = 0.022), confirming RANSAC correctly isolated the same underlying line from the mixed dataset.

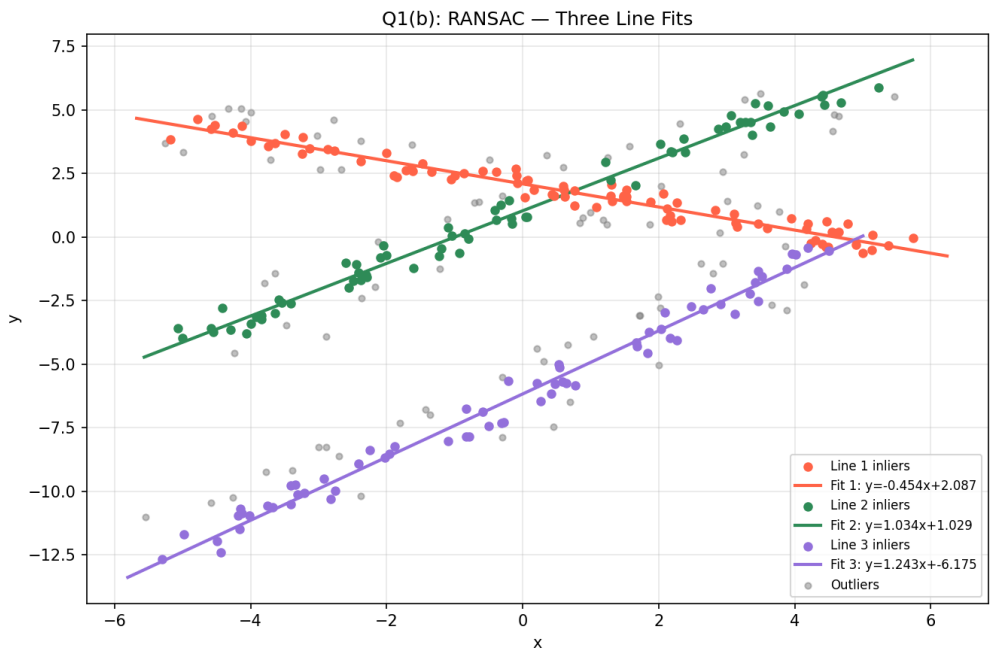


Fig 2: RANSAC - three colour-coded lines with inliers. Grey = outliers.

Q2 - Earring Size Estimation

Given: f = 8 mm, pixel size = 2.2 um, do = 720 mm (camera above flat earring plane).

Thin lens formula: 1/f = 1/do + 1/di => di = (8 x 720)/(720-8) = 8.0899 mm

Magnification: m = di/do = 8.0899/720 = 0.011236

Real-world scale: pixel_size/m = 0.0022/0.011236 = 0.1958 mm/px

Earring outer diameter from bounding box (avg width & height): 389 px => 389 x 0.1958 = 76.2 mm

Both earrings measure 76.2 mm, consistent with large hoop earrings.

```
di_mm = (f_mm * do_mm) / (do_mm - f_mm)
scale_mm_per_px = pixel_mm / (di_mm / do_mm)
```

https://github.com/PathmikaW/cv-assignment02/blob/main/src/q2_earring_size.py

Fig 3: Earring bounding boxes (76.2 mm each)

```
(cv-assignment02) PS D:\MSc\Sem 03\IT5437-Computer Vision\Assignment2\cv-assignment02> uv run python src/q2_earring_size.py
f=8.0 mm,  do=720.0 mm,  di=8.0899 mm
Magnification m = 0.011236
Scale = 0.1958 mm/px

Earring measurements:
  Earring 1: 76.2 mm  (389 px diameter)
  Earring 2: 76.2 mm  (389 px diameter)

Saved: output/q2/q2_earring_size.png
Saved: output/q2/q2_summary.png
```

Terminal output - Q2

Q3 - Circuit Board Homography

Images loaded at 1/4 resolution (IMREAD_REDUCED_COLOR_4). c1 warped to match c2 perspective.

(a) & (b) Manual Correspondences - Homography, Warp and Difference

6 points collected interactively via Listing 1 (listing1_click_points.py - the click tool from the assignment). Landmarks: TL corner hole, USB base, BR/BL corner holes, DC jack, red LED.

```
H_manual, _ = cv2.findHomography(pts1, pts2, method=0)
warped_manual = cv2.warpPerspective(im1, H_manual, (w, h))
diff_manual   = cv2.absdiff(im2, warped_manual)
```

https://github.com/PathmikaW/cv-assignment02/blob/main/src/listing1_click_points.py

H (manual): [[1.0811, -0.0031, -30.59], [-0.0091, 0.9340, -17.26], [0, 0, 1]] | Mean reprojection error: 2.65 px

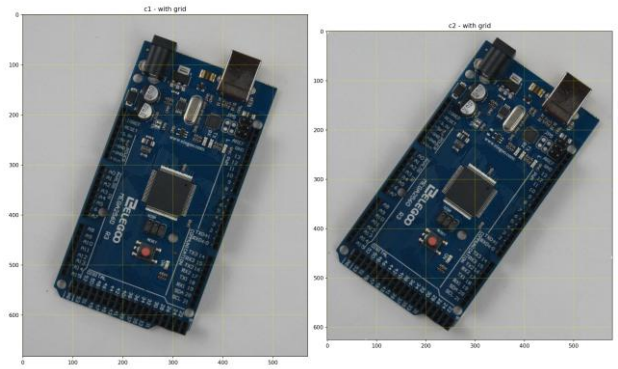


Fig 4: Grid overlay used to identify landmark pixel coordinates in c1 and c2

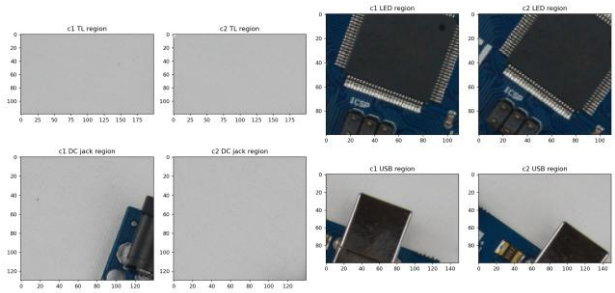


Fig 5: Zoomed landmark regions used to verify clicked point accuracy

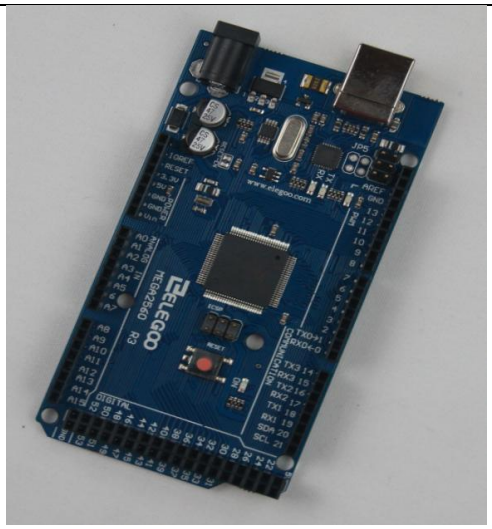


Fig 6: c1 warped (manual)

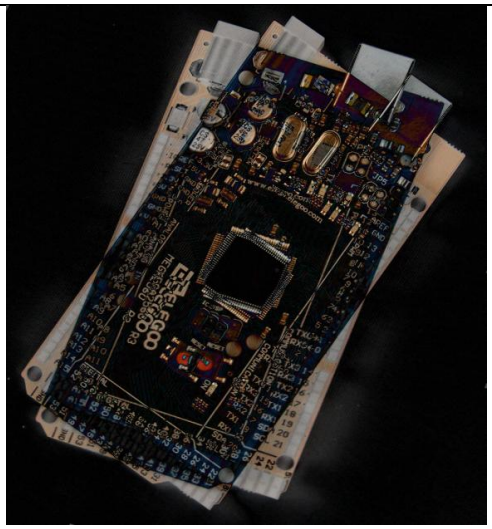


Fig 7: Difference - manual

(c) SIFT Keypoints and Matches

SIFT detects scale/rotation-invariant keypoints. FLANN matcher + Lowe ratio test (0.75) filters ambiguous matches.

Keypoints: 1439 in c1, 1489 in c2. Good matches after ratio test: 1031.

```
sift = cv2.SIFT_create()
good = [m for m, n in flann.knnMatch(des1, des2, k=2) if m.distance < 0.75 * n.distance]
```

https://github.com/PathmikaW/cv-assignment02/blob/main/src/q3_circuit_board.py

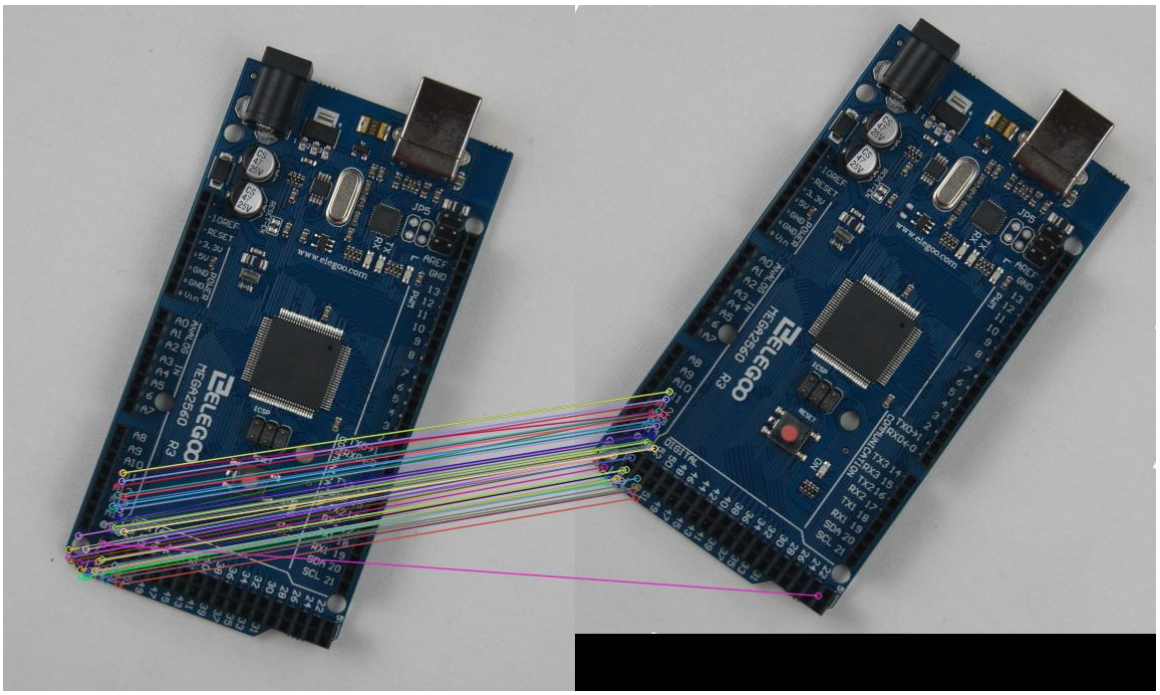


Fig 8: Top 50 SIFT matches between c1 and c2 (1031 good matches total).

(d) SIFT Homography - Warp, Difference and Comparison

1031 matches passed to findHomography with RANSAC (threshold = 5 px).

```
H_sift, mask = cv2.findHomography(src_pts, dst_pts, cv2.RANSAC, 5.0)
```

https://github.com/PathmikaW/cv-assignment02/blob/main/src/q3_circuit_board.py

H (SIFT): [[0.9647, -0.2639, 101.52], [0.2639, 0.9647, -96.05], [0, 0, 1]]

Inliers: 996/1031 (96.6%). H decomposes to: rotation = 15.3 deg, scale = 1.0001.

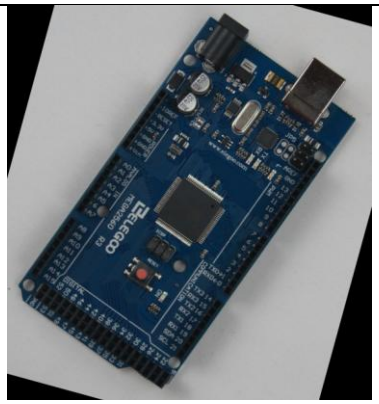


Fig 9: Warped (SIFT)

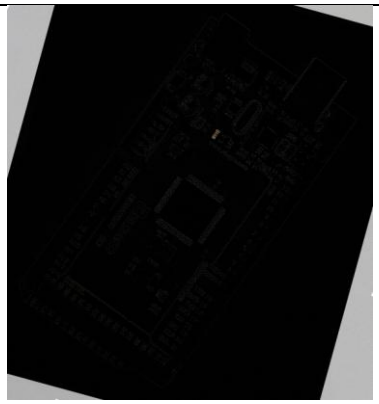
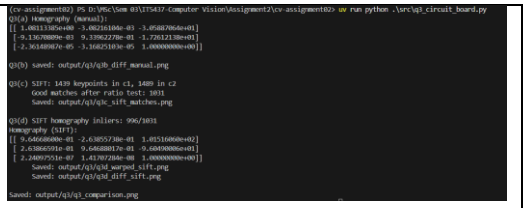


Fig 10: Difference (SIFT)



```
(cv-assignment02) PS D:\MSc\Sem 03\IT5437-Computer Vision\Assignment2\cv-assignment02> uv run python .\src\q3_circuit_board.py
Q3(a) Homography (manual):
[[ 1.08113385e+00 -3.08216104e-03 -3.05887064e+01]
 [-9.13670809e-03  9.33962278e-01 -1.72612138e+01]
 [-2.36148987e-05 -3.10825103e-05  1.00000000e+00]]

Q3(b) saved: output/q3/q3b_diff_manual.png

Q3(c) SIFT: 1439 keypoints in c1, 1489 in c2
      Good matches after ratio test: 1031
      Saved: output/q3/q3c_sift_matches.png

Q3(d) SIFT homography inliers: 996/1031
Homography (SIFT):
[[ 9.64668600e-01 -2.63855738e-01  1.05516060e+02]
 [ 2.63866591e-01  9.64688017e-01 -9.60490006e+01]
 [ 2.24897551e-07  1.41787204e-08  1.00000000e+00]]
      Saved: output/q3/q3d_warped_sift.png
      Saved: output/q3/q3d_diff_sift.png

Saved: output/q3/q3_comparison.png
```

Terminal output - Q3

Comparison - Manual vs SIFT: The manual approach uses only 6 clicked points with inherent human imprecision (~2-8 px error), leaving visible misalignment in the difference image - bright patches at board edges, USB connector, and the IC chip. SIFT uses 996 inlier correspondences spread across the entire board, giving a much more constrained homography. The SIFT difference image is nearly black, confirming both images show the same board (as stated in the assignment). SIFT is fully automatic and handles scale/rotation naturally. Its main limitation is that it requires sufficient image texture - on featureless surfaces, manual selection would be the only option.